Lab Assignment #4a

Math 437 - Modern Data Analysis - Jason Soto

Introduction

In a 2020 article, Murray et al. attempted to understand the relationship between a professor's research output and their perceived teaching effectiveness.

To quantify perceived teaching effectiveness, they scraped Rate My Professor and calculated an "overall effectiveness" score (based on the data dictionary, this appears to just be the average star rating, on a scale from 1-5, from reviews posted between Spring 2010 and Fall 2017 semesters).

An anonymized, processed dataset and accompanying data dictionary is publicly available; I've removed most of the research-related variables to shrink the size of the dataset and shared it with you on Canvas as *RateMyProfessor*.

```
here::i_am("Labs/Lab04a.qmd")

# load packages
library(ISLR2)
library(car)

# load data
RateMyProfessor <- readr::read_csv(here::here("Data/RateMyProfessor.csv"))
```

In this and the next lab assignment, we will build increasingly complex models to understand what goes into a professor's RateMyProfessor rating.

This lab assignment is worth a total of **15 points**.

Problem 1: Simple Linear Regression (6 pts total)

Part a (Code: 0.5 pts)

Run the code in ISLR Lab 3.6.2. Put each chunk from the textbook in its own chunk.

Solution

```r
head(Boston)
```

```
     crim zn indus chas   nox    rm  age    dis rad tax ptratio lstat medv
1 0.00632 18  2.31    0 0.538 6.575 65.2 4.0900   1 296    15.3  4.98 24.0
2 0.02731  0  7.07    0 0.469 6.421 78.9 4.9671   2 242    17.8  9.14 21.6
3 0.02729  0  7.07    0 0.469 7.185 61.1 4.9671   2 242    17.8  4.03 34.7
4 0.03237  0  2.18    0 0.458 6.998 45.8 6.0622   3 222    18.7  2.94 33.4
5 0.06905  0  2.18    0 0.458 7.147 54.2 6.0622   3 222    18.7  5.33 36.2
6 0.02985  0  2.18    0 0.458 6.430 58.7 6.0622   3 222    18.7  5.21 28.7
```

```r
attach(Boston)
lm.fit <- lm(medv ~ lstat)
```

```r
lm.fit <- lm(medv ~ lstat, data = Boston)
```

```r
lm.fit
```

```
Call:
lm(formula = medv ~ lstat, data = Boston)

Coefficients:
(Intercept)        lstat
      34.55        -0.95
```

```r
summary(lm.fit)
```

```
Call:
lm(formula = medv ~ lstat, data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max
-15.168  -3.990  -1.318   2.034  24.500

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 34.55384    0.56263   61.41   <2e-16 ***
```

```
lstat       -0.95005    0.03873  -24.53   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.216 on 504 degrees of freedom
Multiple R-squared:  0.5441,    Adjusted R-squared:  0.5432
F-statistic: 601.6 on 1 and 504 DF,  p-value: < 2.2e-16
```

```r
names(lm.fit)
```

```
 [1] "coefficients"  "residuals"     "effects"       "rank"
 [5] "fitted.values" "assign"        "qr"            "df.residual"
 [9] "xlevels"       "call"          "terms"         "model"
```

```r
coef(lm.fit)      #safer than data$coefficients
```

```
(Intercept)       lstat
 34.5538409  -0.9500494
```

```r
confint(lm.fit)
```

```
                2.5 %     97.5 %
(Intercept) 33.448457 35.6592247
lstat       -1.026148 -0.8739505
```

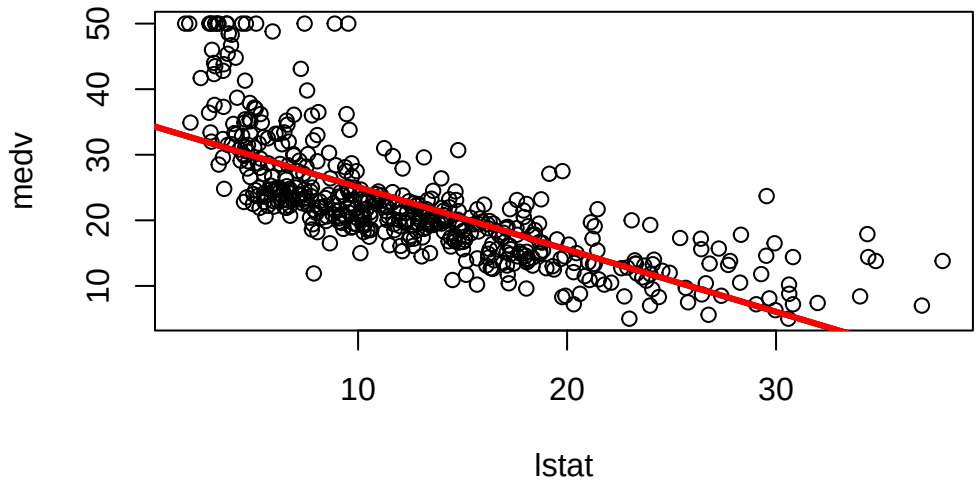
```r
predict(lm.fit, data.frame(lstat = (c(5, 10, 15))), interval = "confidence")
```

```
       fit      lwr      upr
1 29.80359 29.00741 30.59978
2 25.05335 24.47413 25.63256
3 20.30310 19.73159 20.87461
```

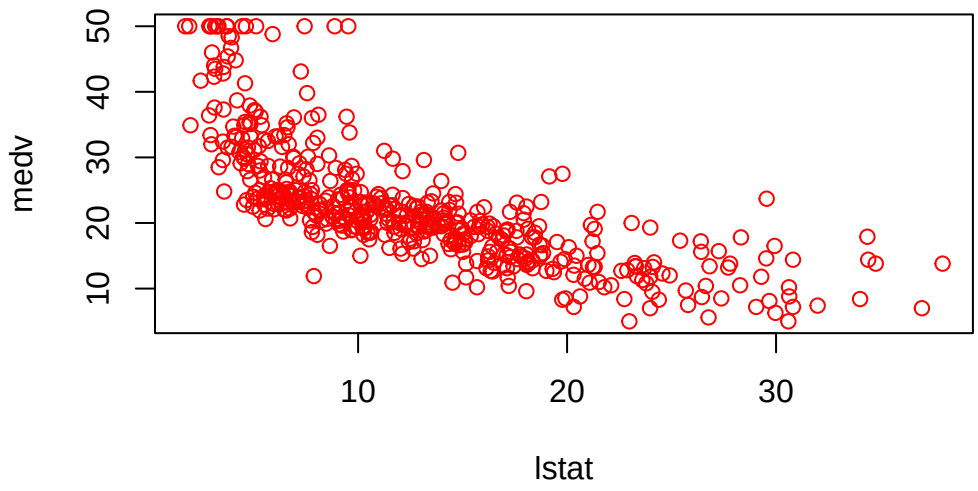
```r
predict(lm.fit, data.frame(lstat = (c(5, 10, 15))), interval = "prediction")
```

```
       fit       lwr      upr
1 29.80359 17.565675 42.04151
2 25.05335 12.827626 37.27907
3 20.30310  8.077742 32.52846
```

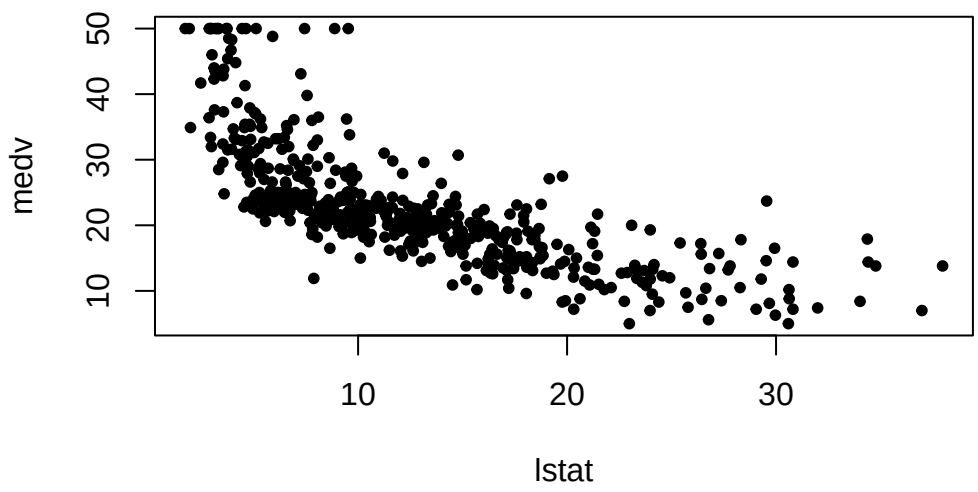
```r
plot(lstat, medv)
abline(lm.fit, lwd = 3)
abline(lm.fit, lwd = 3, col = "red")
```



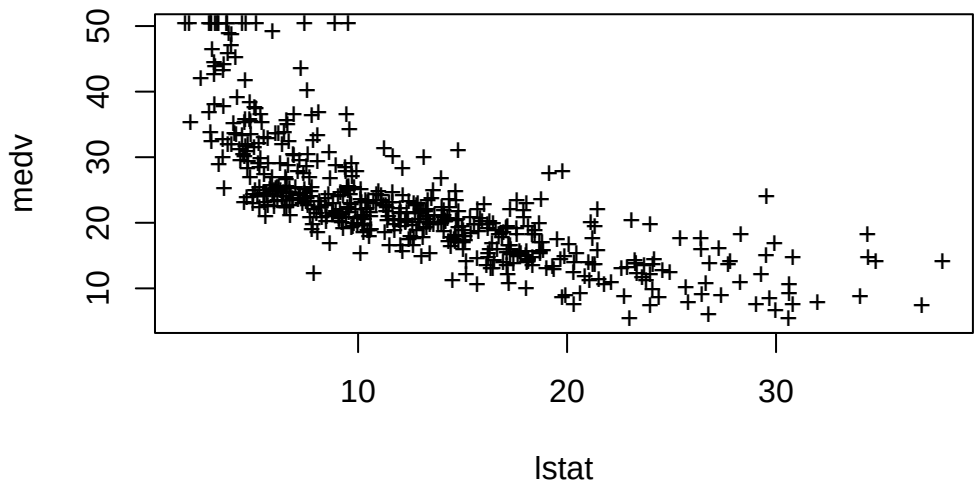

```r
plot(lstat, medv, col = "red")
```



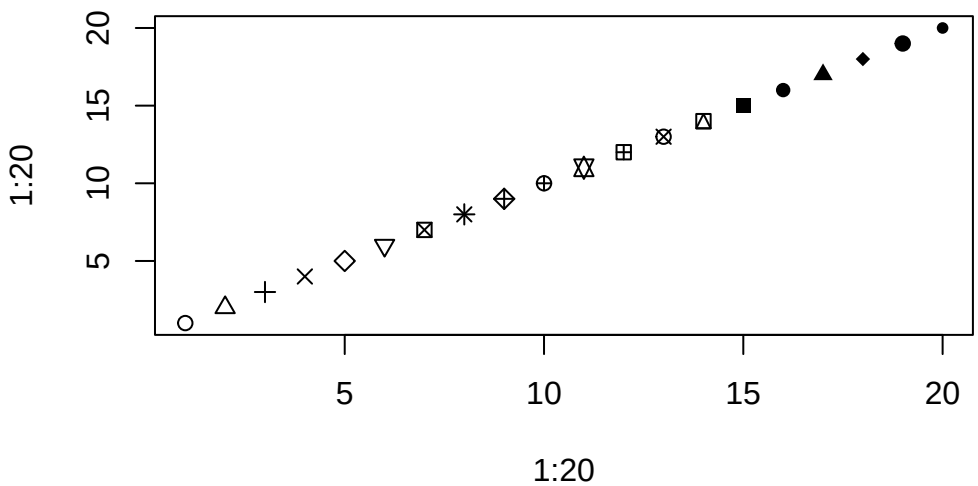

```r
plot(lstat, medv, pch = 20)
```



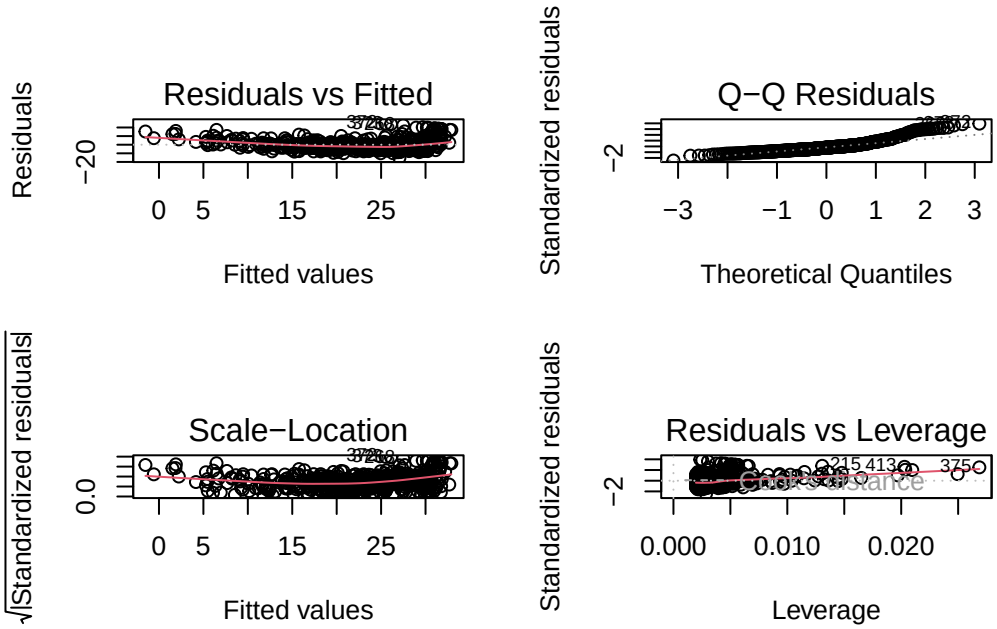
```r
plot(lstat, medv, pch = "+")
```



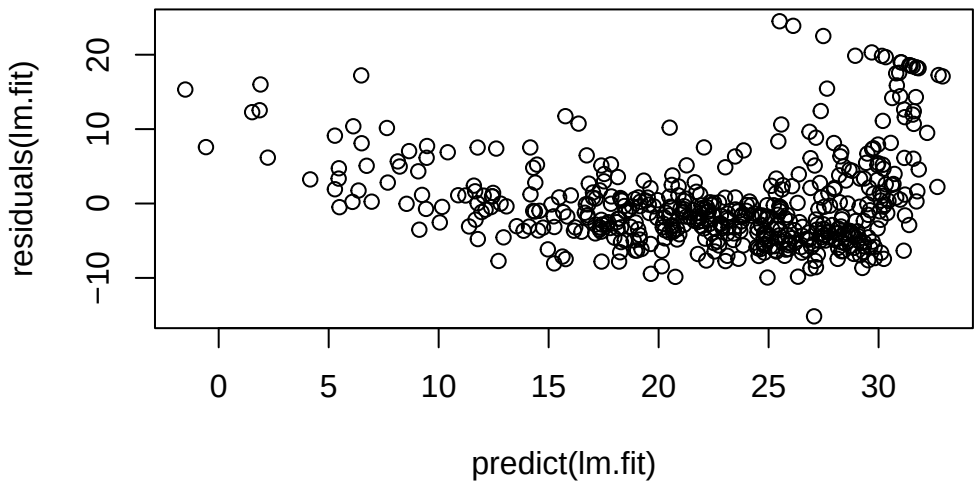

```r
plot(1:20, 1:20, pch = 1:20)
```



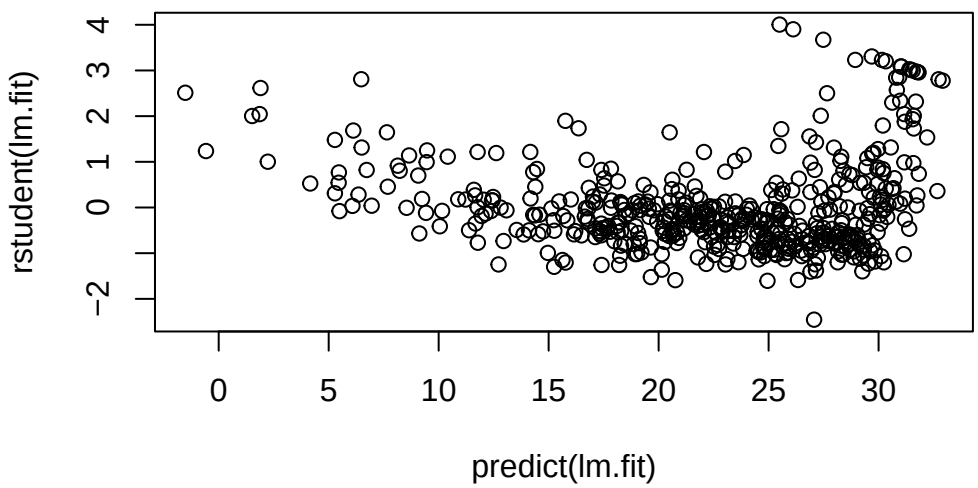
```r
par(mfrow = c(2, 2))
plot(lm.fit)
```



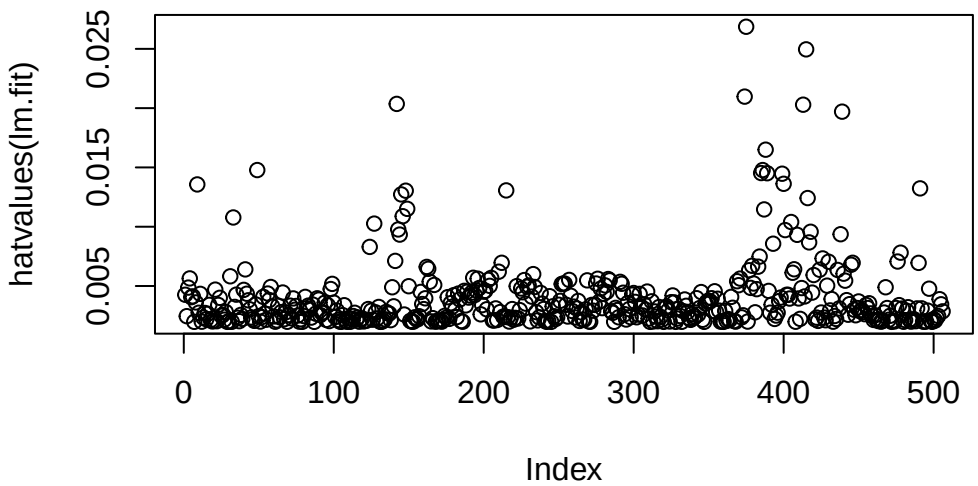

```r
plot(predict(lm.fit), residuals(lm.fit))
```




```
plot(predict(lm.fit), rstudent(lm.fit))
```



```r
plot(hatvalues(lm.fit))
```




```r
which.max(hatvalues(lm.fit))
```

```
375
375
```

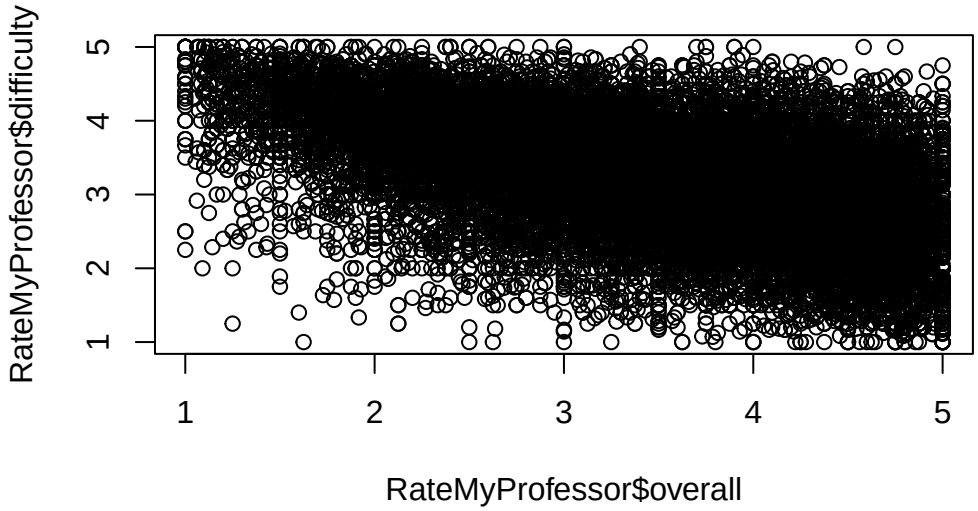
Part b (Code: 0.5 pts; Explanation: 1 pt)

Let's start our investigation of the RateMyProfessor data by looking at the relationship between `overall` and `difficulty`, the average rating for professor difficulty (also on a 1-5 scale with this time 1 = easiest, 5 = hardest). Create a scatterplot to visualize this relationship. For now, don't worry about making it look nice or fixing labels.

Write a short paragraph describing what you can tell from the scatterplot about `overall`, `difficulty`, and/or the relationship between the two variables. Aim for roughly 2-4 sentences and assume that the person reading your paragraph knows no statistical jargon (e.g., instead of saying "strong linear relationship", say something like "the points follow very close to a line").

Solution

```
plot(RateMyProfessor$overall, RateMyProfessor$difficulty)
```



What I can tell about this data from the plot is that the easier the professor is, the better their review becomes, though it still varies in the right half, there are still many difficult professors with good reviews, so it funnels, with the little end on the left. We see a negative linear pattern. Long story short, good reviews will very, but the tougher you are, the chance of a worse review is more likely.

Part c (Code: 0.5 pts)

Fit a simple linear regression model predicting `overall` from `difficulty`. Obtain the `summary` of the model and use it to write the equation of the least-squares line.

Solution

```
lm_simple <- lm(overall ~ difficulty, data = RateMyProfessor)
summary(lm.fit)
```

```
Call:
lm(formula = medv ~ lstat, data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max 
-15.168  -3.990  -1.318   2.034  24.500 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 34.55384    0.56263   61.41   <2e-16 ***
lstat       -0.95005    0.03873  -24.53   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.216 on 504 degrees of freedom
Multiple R-squared:  0.5441,	Adjusted R-squared:  0.5432 
F-statistic: 601.6 on 1 and 504 DF,  p-value: < 2.2e-16
```

yhat = 5.5617 - .6412(difficulty)

Part d (Explanation: 1 pt)

Does the slope of the regression model make real-world sense with this data? If so, explain how to interpret it. If not, explain why not.

Solution

The slope does makes sense, a sort of resentment grows when a class or professor is difficult, so the slope of the difficulty should be negative, slowly bringing down the review of the professor based on difficulty.

Part e (Explanation: 1 pt)

Does the intercept of the regression model make real-world sense with this data? If so, explain how to interpret it. If not, explain why not.

Solution

Kind of? The lowest difficulty is 1, so the highest possible score would technically be less than 5, the highest possible grade, and while reviews can match that, maybe true professor ranking can't? Not sure but the current regression model should be good enough to work.

Problem 2: Multiple Linear Regression (3.5 pts total)

Part a (Code: 0.5 pts)

Run the code in ISLR Lab 3.6.3. Put each chunk from the textbook in its own chunk.

Solution

```r
lm.fit <- lm(medv ~ lstat + age, data = Boston)
summary(lm.fit)
```

```

Call:
lm(formula = medv ~ lstat + age, data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max 
-15.981  -3.978  -1.283   1.968  23.158 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 33.22276    0.73085  45.458  < 2e-16 ***
lstat       -1.03207    0.04819 -21.416  < 2e-16 ***
age          0.03454    0.01223   2.826  0.00491 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.173 on 503 degrees of freedom
Multiple R-squared:  0.5513,	Adjusted R-squared:  0.5495 
F-statistic:   309 on 2 and 503 DF,  p-value: < 2.2e-16
```

```r
lm.fit <- lm(medv ~ ., data = Boston)
summary(lm.fit)
```

```
Call:
lm(formula = medv ~ ., data = Boston)

Residuals:
     Min       1Q   Median       3Q      Max 
-15.1304  -2.7673  -0.5814   1.9414  26.2526 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  41.617270   4.936039   8.431 3.79e-16 ***
crim         -0.121389   0.033000  -3.678 0.000261 ***
zn            0.046963   0.013879   3.384 0.000772 ***
indus         0.013468   0.062145   0.217 0.828520    
chas          2.839993   0.870007   3.264 0.001173 ** 
nox         -18.758022   3.851355  -4.870 1.50e-06 ***
rm            3.658119   0.420246   8.705  < 2e-16 ***
age           0.003611   0.013329   0.271 0.786595    
dis          -1.490754   0.201623  -7.394 6.17e-13 ***
rad           0.289405   0.066908   4.325 1.84e-05 ***
tax          -0.012682   0.003801  -3.337 0.000912 ***
ptratio      -0.937533   0.132206  -7.091 4.63e-12 ***
lstat        -0.552019   0.050659 -10.897  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.798 on 493 degrees of freedom
Multiple R-squared:  0.7343,	Adjusted R-squared:  0.7278 
F-statistic: 113.5 on 12 and 493 DF,  p-value: < 2.2e-16
```

```r
library(car)
vif(lm.fit)
```

```
    crim       zn    indus     chas      nox       rm      age      dis 
1.767486 2.298459 3.987181 1.071168 4.369093 1.912532 3.088232 3.954037 
     rad      tax  ptratio    lstat 
7.445301 9.002158 1.797060 2.870777 
```

```r
lm.fit1 <- lm(medv ~ . - age, data = Boston) # can do this instead but this lowkirkenuienly l
summary(lm.fit1)
```

```
Call:
lm(formula = medv ~ . - age, data = Boston)

Residuals:
     Min       1Q   Median       3Q      Max 
-15.1851  -2.7330  -0.6116   1.8555  26.3838 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  41.525128   4.919684   8.441 3.52e-16 ***
crim         -0.121426   0.032969  -3.683 0.000256 ***
zn            0.046512   0.013766   3.379 0.000785 ***
indus         0.013451   0.062086   0.217 0.828577    
chas          2.852773   0.867912   3.287 0.001085 ** 
nox         -18.485070   3.713714  -4.978 8.91e-07 ***
rm            3.681070   0.411230   8.951  < 2e-16 ***
dis          -1.506777   0.192570  -7.825 3.12e-14 ***
rad           0.287940   0.066627   4.322 1.87e-05 ***
tax          -0.012653   0.003796  -3.333 0.000923 ***
ptratio      -0.934649   0.131653  -7.099 4.39e-12 ***
lstat        -0.547409   0.047669 -11.483  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.794 on 494 degrees of freedom
Multiple R-squared:  0.7343,	Adjusted R-squared:  0.7284 
F-statistic: 124.1 on 11 and 494 DF,  p-value: < 2.2e-16
```

Part b (Code: 0.5 pts)

The `interest` variable represents the average student interest in the subject of the course (1 = "Low" interest; 5 = "It's My Life"). Note that this rating no longer exists on the website.

Fit a multiple linear regression model predicting `overall` from `difficulty` and `interest`. Obtain the `summary` of the model and use it to write the equation of the least-squares line.

Solution

```r
lm_indicators <- lm(
  overall ~ difficulty + interest,
  data = RateMyProfessor
)
```

```r
summary(lm_indicators)
```

```
Call:
lm(formula = overall ~ difficulty + interest, data = RateMyProfessor)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.5532 -0.4654  0.0446  0.4942  2.5945 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  3.440577   0.034964   98.40   <2e-16 ***
difficulty  -0.530432   0.006347  -83.57   <2e-16 ***
interest     0.517124   0.006961   74.29   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7071 on 22035 degrees of freedom
Multiple R-squared:  0.4255,	Adjusted R-squared:  0.4254 
F-statistic:  8160 on 2 and 22035 DF,  p-value: < 2.2e-16
```

yhat = 3.4406 - .5304(difficulty) + .5171(interest)

Part c (Explanation: 1 pt)

Notice that the slope corresponding to the `difficulty` variable has a different value in the simple linear regression model and the multiple linear regression model from Part b.

How should we interpret the slope corresponding to `difficulty` in this new model?

Solution

I believe this can be interpreted the same way, the reason for the less negatively effecting slope value though is because we added interest into the equation, adding more context, voice, and power to yhat's final value.

Part d (Code: 0.5 pts; Explanation: 1 pt)

Does your interpretation from Part c actually make sense given the collected data? Support your answer by using the `vif` function and interpreting the output.

Solution

```r
vif(lm_indicators)
```

```
difficulty   interest
  1.058407   1.058407
```

I assume my interpretation is correct, considering how well they work with each other, having vif values very close to 1

Problem 3: Indicator Variables

Part a (Code: 0.5 pts)

Run the code in ISLR Lab 3.6.6. Put each chunk from the textbook in its own chunk.

Solution

```r
head(Carseats)
```

```
  Sales CompPrice Income Advertising Population Price ShelveLoc Age Education
1  9.50       138     73          11        276   120       Bad  42        17
2 11.22       111     48          16        260    83      Good  65        10
3 10.06       113     35          10        269    80    Medium  59        12
4  7.40       117    100           4        466    97    Medium  55        14
5  4.15       141     64           3        340   128       Bad  38        13
6 10.81       124    113          13        501    72       Bad  78        16
  Urban  US
1   Yes Yes
2   Yes Yes
3   Yes Yes
4   Yes Yes
5   Yes  No
6    No Yes
```

```r
lm.fit <- lm(Sales ~ . + Income:Advertising + Price:Age, data = Carseats)

summary(lm.fit)
```

```
Call:
lm(formula = Sales ~ . + Income:Advertising + Price:Age, data = Carseats)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.9208 -0.7503  0.0177  0.6754  3.3413 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)         6.5755654  1.0087470   6.519 2.22e-10 ***
CompPrice           0.0929371  0.0041183  22.567  < 2e-16 ***
Income              0.0108940  0.0026044   4.183 3.57e-05 ***
Advertising         0.0702462  0.0226091   3.107 0.002030 ** 
Population          0.0001592  0.0003679   0.433 0.665330    
Price              -0.1008064  0.0074399 -13.549  < 2e-16 ***
ShelveLocGood       4.8486762  0.1528378  31.724  < 2e-16 ***
ShelveLocMedium     1.9532620  0.1257682  15.531  < 2e-16 ***
Age                -0.0579466  0.0159506  -3.633 0.000318 ***
Education          -0.0208525  0.0196131  -1.063 0.288361    
UrbanYes            0.1401597  0.1124019   1.247 0.213171    
USYes              -0.1575571  0.1489234  -1.058 0.290729    
Income:Advertising  0.0007510  0.0002784   2.698 0.007290 ** 
Price:Age           0.0001068  0.0001333   0.801 0.423812    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.011 on 386 degrees of freedom
Multiple R-squared:  0.8761,	Adjusted R-squared:  0.8719 
F-statistic:   210 on 13 and 386 DF,  p-value: < 2.2e-16
```

```r
attach(Carseats)

contrasts(ShelveLoc)
```

```
       Good Medium
Bad       0      0
Good      1      0
Medium    0      1
```

Part b (Code: 0.5 pts)

Using the RateMyProfessor dataset, fit a linear model predicting the overall rating of a professor (`overall`) from the difficulty rating (`difficulty`), chili pepper rating (`hotness`), and rank (`rank`).

Obtain the `summary` of the model and use it to write the equation of the least-squares line.

Solution

```r
mlm <- lm(overall ~ difficulty + hotness + rank, data = RateMyProfessor)
summary(mlm)
```

```
Call:
lm(formula = overall ~ difficulty + hotness + rank, data = RateMyProfessor)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.3163 -0.5038  0.0385  0.5274  2.4384 

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)    
(Intercept)              5.301027   0.025082 211.351  < 2e-16 ***
difficulty              -0.587803   0.006605 -88.997  < 2e-16 ***
hotnesshot               0.636429   0.012819  49.649  < 2e-16 ***
rankAssociate Professor -0.050371   0.015088  -3.338 0.000844 ***
rankProfessor           -0.047455   0.014747  -3.218 0.001293 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7475 on 22033 degrees of freedom
Multiple R-squared:  0.3581,	Adjusted R-squared:  0.3579 
F-statistic:  3072 on 4 and 22033 DF,  p-value: < 2.2e-16
```

yhat = 5.301 - .5878(difficulty) + .6364(hotness == "hot") - .0503(rank == "Associate Professor") - .0475(rank == "Professor")

Part c (Explanation: 1.5 pts)

What are the reference levels for each categorical variable in the model from Part c? How do you know?

Solution

The reference level for hotness is "not hot" since the coefficient is given when professor is "hot". The next reference level is for rank, that is "Assistant Professor", using the same logic. That should be it since any other variable has a coefficient that it can affect.

Part d (Explanation: 2 pts)

Holding difficulty constant, which of the following instructors would be predicted to have the highest overall rating? Which would be predicted to have the lowest overall rating? Explain your reasoning.

- Attractive Assistant Professor
- Attractive Associate Professor
- Attractive Professor
- Less-attractive Assistant Professor
- Less-attractive Associate Professor
- Less-attractive Professor

Solution

The highest overall rating should be the Assistant Professor. This because other 2 ranks have negative effects, while assistant has none at all in our least square lines. Attractive has positive effect, while chopped has none.

The lowest overall rating should be the Less-Attractive Associate Professor. Using the same idea, no positive effect is given due to looks. Assistant professor also has the most negative impact due to its coefficient.